and accelerated motion—using the precision of twenty-first-century rocket science—you are able to stave off disaster.

The recognition that gravity and accelerated motion are profoundly interwoven is the key insight that Einstein had one happy day in the Bern patent office. Although the bomb experience highlights the essence of his idea, it is worth rephrasing it in a framework closer to that of Chapter 2. For this purpose, recall that if you are put into a sealed, windowless compartment that is not accelerating, there is no way for you to determine your speed. The compartment looks the same and any experiments you do yield identical results regardless of how fast you are moving. More fundamentally, without outside benchmarks for comparison there is no way that a velocity can even be assigned to your state of motion. On the other hand, if you are accelerating, then even with your perceptions limited to the confines of your sealed compartment, you will feel a force on your body. For instance, if your forward-facing chair is bolted to the floor and your compartment is being accelerated forward, you will feel the force of your seat on your back just as with the car described by Albert. Similarly, if your compartment is being accelerated upward you will feel the force of the floor on your feet. Einstein's realization was that within the confines of your tiny compartment, you will not be able to distinguish these accelerated situations from ones without acceleration but with gravity: When their magnitudes are judiciously adjusted, the force you feel from a gravitational field or from accelerated motion are indistinguishable. If your compartment is placidly sitting upright on the earth's surface, you will feel the familiar force of the floor on your feet, just as in the scenario of upward acceleration; this is exactly the same equivalence Albert exploited in his solution for launching the terrorist bomb into space. If your compartment is resting on its back end you will feel the force of your seat on your back (preventing you from falling), just as when you were accelerating horizontally. Einstein called the indistinguishability between accelerated motion and gravity the equivalence principle.

2

This description shows that general relativity finishes a job initiated by special relativity. Through its principle of relativity, the special theory of relativity declares a democracy of observational